

Eclipse IDE

Java IDE with Git, Maven/Gradle, debugging and test execution.

What it is

Eclipse IDE is an extensible development environment widely used in the Java ecosystem. QA Automation engineers can import projects, run JUnit/TestNG, debug tests and work with Maven, Gradle and Git.

Prerequisites

Use a JDK version supported by the project. Eclipse may bundle a runtime for the IDE, but your Java project should still use the JDK required by its stack.

Windows installation

Download Eclipse Installer from the official site, run it, choose Eclipse IDE for Java Developers or Enterprise Java and Web Developers, select the install folder and finish. Then choose a workspace.

macOS and Linux

Download the package matching your OS and architecture and follow the Eclipse Foundation installer/package instructions.

QA setup

Configure the JDK under Installed JREs, import the Maven/Gradle project, resolve dependencies, configure Git and run a test class. Use breakpoints and Debug to investigate failures.

Validation and troubleshooting

Check project JDK, encoding, corporate proxy, Maven/Gradle and dependency resolution. If compilation fails, validate Java and the build outside the IDE first.

QA tip

Always connect the practice to product risk and the evidence needed for quality decisions.

Official resources

Site oficial: <https://www.eclipse.org/>

Download / instalação: <https://www.eclipse.org/downloads/>

Documentação: <https://help.eclipse.org/>